

TRẦN MẠNH KHANG

AI Engineer — LLM, AI Agents & Model Evaluation

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SUMMARY

Final-year Software Engineering student focused on making LLM systems measurably reliable. Built and fine-tuned an LLM workflow for customer-call quality assessment at Concentrix, and ran a 10-phase independent study on whether a small LLM can catch its own mistakes, using automatic answer checking instead of an AI judge. Comfortable turning failures into controlled experiments, reporting negative results honestly, and building agents with safety guardrails.

SELECTED PROJECTS

LLM Self-Correction Research

Personal research project · 10 phases

Problem: can a small language model detect and fix its own wrong answers?

- **Evaluation without an AI judge:** math answers checked symbolically and code run against tests; all test sets frozen and kept separate from training data.
- **Voting works:** taking the majority over 5 independent attempts raised GSM8K accuracy from 75.0% to 83.25–85.5% on 400 fresh problems across three model versions ($p < 0.001$).
- **Knows that, not which:** disagreement between two attempts caught 82–85% of wrong first answers, but the model picked the right one of two conflicting solutions only 44–52% of the time.
- **Fixes that failed, reported as such:** a fine-tuned judge (808 balanced examples) stayed at chance (52.1% → 50.7%) and trailed voting by 6 points; preference tuning raised KEEP recall 81% → 85% without better error detection.
- **Blind re-solve helps:** a second attempt that does not see the earlier answer fixed far more errors; routing flagged answers to it lifted accuracy from 70.75% to 74.5–77.5%. All data, configs and evaluation code kept reproducible.

Ciel — Personal AI Agent

Personal project · Python, Docker

Problem: let an LLM plan and act with real tools without unsafe or duplicate actions.

- **Brain/Worker design:** one model classifies intent and plans, another responds; deterministic code validates plans, permissions, budgets and cancellation, so the model proposes and the code enforces.
- **Guardrails:** malformed multi-tool plans stop before execution, risky actions need confirmation, file access is sandboxed, and outgoing messages and reminders are deduplicated across restarts.
- **Product features:** monthly/weekly planning, one-time reminders, long-term memory (ChromaDB) and proactive Telegram notifications on one runtime shared by CLI, API/web UI and Telegram; deployed with Docker.
- **Prompt-improvement tool:** mines repeated failures and may edit only one allow-listed prompt string; auto-reverts if unit tests fail. Backed by a regression test suite.

EXPERIENCE

AI Engineer Intern — Concentrix, Vietnam

Mar 2026 – Jul 2026

- Built an LLM-based workflow to assess customer-call quality: converted conversations into structured data and validated predictions against human-labeled samples.
- Fine-tuned a small open-source LLM, focusing training on hard cases where earlier versions failed, which substantially improved accuracy.
- Benchmarked it against leading commercial models: comparable on straightforward cases, stronger on difficult ones, at a lower inference cost.
- Tested AI agent workflows for reliability: retry/fallback behavior, tool-response validation, edge-case documentation and prompt improvements.

Exchange Student Program — JeonJu University, Korea

2024

- Studied AI, Machine Learning and Deep Learning as an exchange student.

EDUCATION

B.Eng. Software Engineering — Ton Duc Thang University

2021 – 2026

IELTS 6.0

SKILLS

LLM & ML: PyTorch, Unsloth, fine-tuning (SFT, DPO, GRPO, ORPO), model evaluation, representation probing, activation steering

Agent systems: LangChain, LangGraph, tool calling, retry/fallback workflows, guardrails, ChromaDB (RAG)

Computer vision: YOLOv8, OpenCV

Software & backend: Python, Java, JavaScript, TypeScript, Node.js, Express, REST APIs, MongoDB, Docker, vLLM, Git